SIEM LOG MONITORING AND THREAT DETECTION USING SPLUNK AND SYSMON

INTRODUCTION:

The growing rate of cyber threats has made real-time monitoring and detection essential in modern computing environments. Security Information and Event Management (SIEM) systems play a critical role in collecting, analyzing, and correlating log data from multiple sources to identify suspicious activities. This project presents the development of a SIEM lab using Splunk Enterprise and Sysmon to monitor, detect, and analyze simulated cyberattacks on a Windows system. The study focuses on practical detection of brute force login attempts, PowerShell-based attacks, and suspicious process activities.

OBJECTIVES:

The objectives of this project include:

- To configure a functional SIEM environment
- To collect and analyze Windows and Sysmon logs
- To simulate real-world cyberattacks
- To develop detection rules using Splunk
- To visualize security events through dashboards

TOOLS AND TECHNOLOGIES:

- Splunk Enterprise
- Splunk Universal Forwarder
- Sysmon
- Windows Event Viewer
- Splunk Processing Language (SPL)

SYSTEM ARCHITECTURE:

The system follows a centralized log collection architecture:

Sysmon generates detailed endpoint logs, which are stored in Windows Event Logs. These logs are forwarded using Splunk Universal Forwarder to Splunk Enterprise, where they are indexed, analyzed, and visualized through dashboards.

METHODOLOGY:

Log Collection

Sysmon was installed and configured to capture system activities such as process creation, network connections, and file modifications. The Splunk Universal Forwarder was configured to collect logs from:

- Security logs
- System logs
- Application logs
- Sysmon operational logs

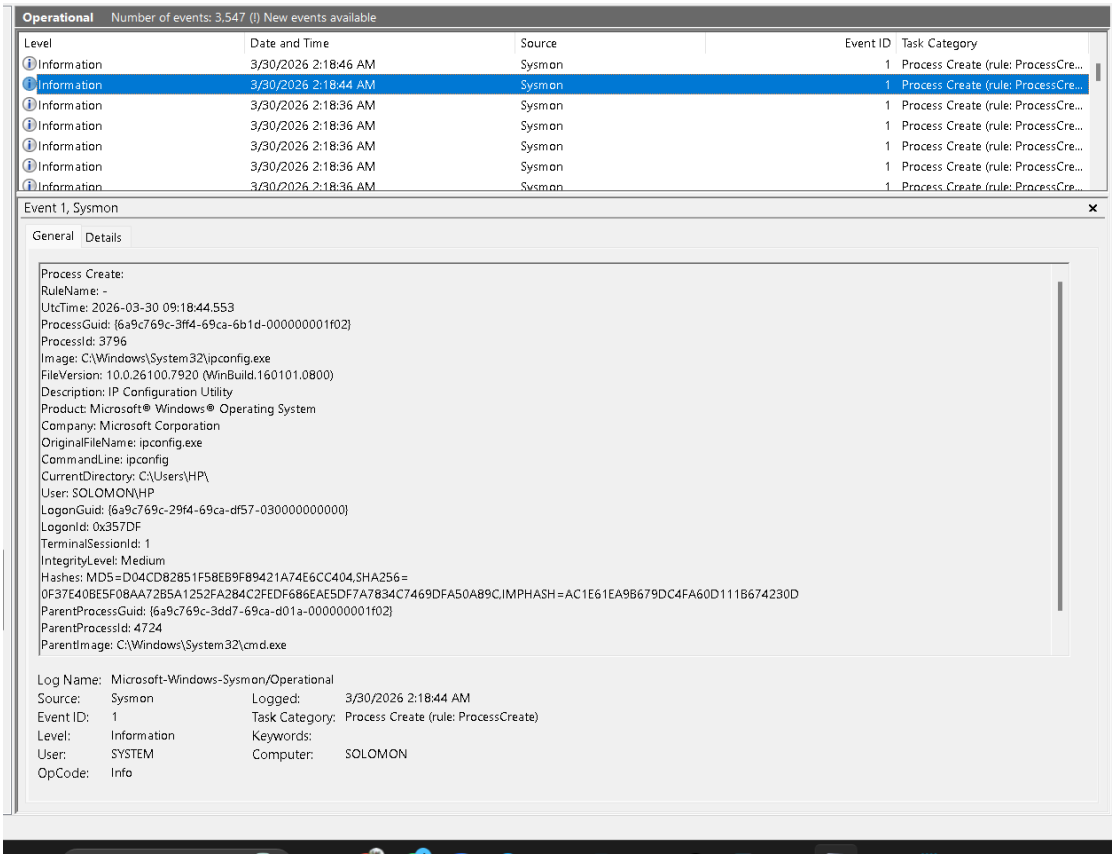


Figure 1: Sysmon Operational Logs Capturing System Activities

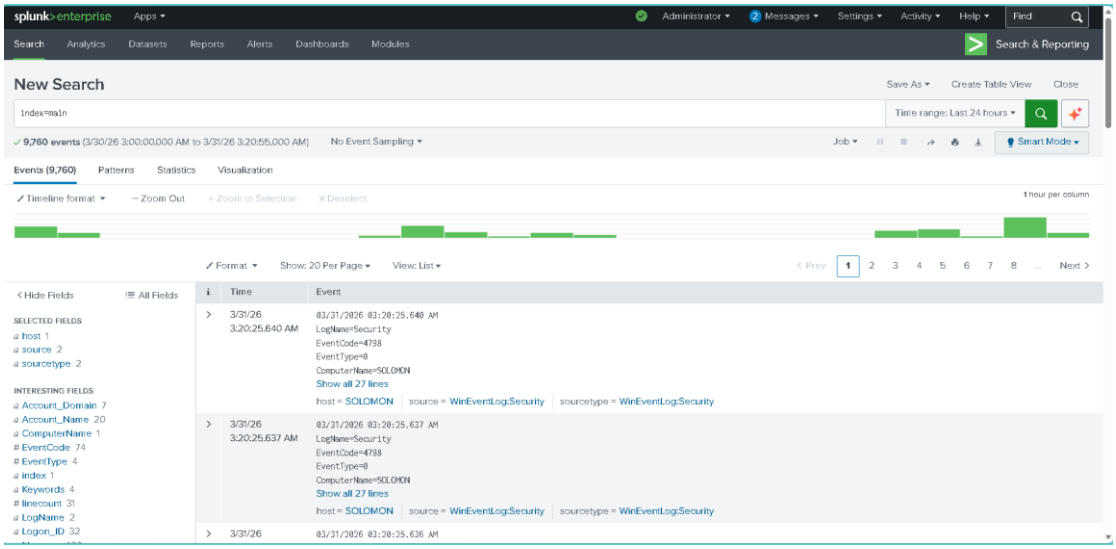


Figure 2: Logs Successfully Ingested into Splunk SIEM

Attack Simulation

The following attack scenarios were simulated:

a. Brute Force Attack

Repeated failed login attempts were generated using incorrect credentials to simulate unauthorized access attempts.

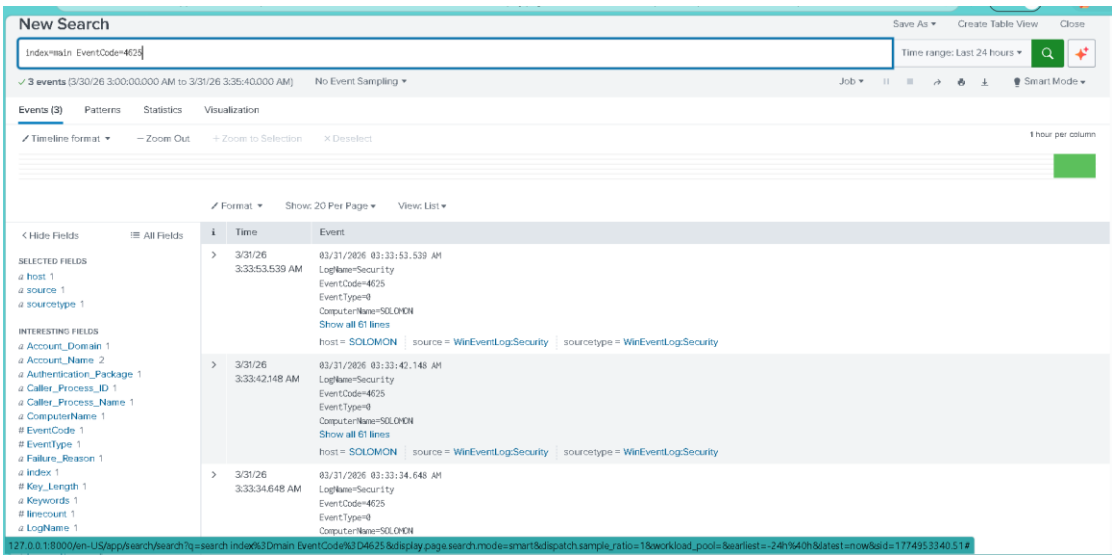
```
C:\Users\HP>runas /user:User1 cmd
Enter the password for User1:
Attempting to start cmd as user "SOLOMON\User1" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\HP>runas /user:User3 cmd
Enter the password for User3:
Attempting to start cmd as user "SOLOMON\User3" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\HP>runas /user:User4 cmd
Enter the password for User4:
Attempting to start cmd as user "SOLOMON\User4" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\HP>runas /user:ubong cmd
Enter the password for ubong:
Attempting to start cmd as user "SOLOMON\ubong" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
```

Figure 3: Simulation of Brute Force Attack Using Multiple Failed Logins



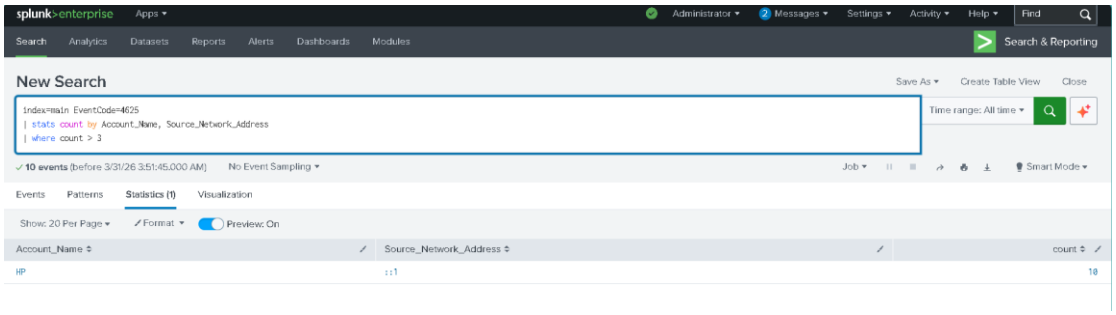


Figure 4: Detection of Failed Login Attempts in Splunk

b. PowerShell Attack

Malicious PowerShell commands were executed using execution policy bypass and remote script loading techniques.

```
>> PS C:\Users\HP> Set-ItemProperty -Path "HKLM:\Software\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Name "EnableScriptBlockLogging" -Value 1
>> Set-ItemProperty : Requested registry access is not allowed.
>> At line:1 char:1
>> + Set-ItemProperty -Path "HKLM:\Software\Policies\Microsoft\Windows\Pow ...
At line:1 char:2
+ + Set-ItemProperty -Path "HKLM:\Software\Policies\Microsoft\Windows\P ...
+  ~
Missing expression after unary operator '+'.
At line:1 char:3
+ + Set-ItemProperty -Path "HKLM:\Software\Policies\Microsoft\Windows\P ...
+   ~~~~~~~~~~~~~~~~
Unexpected token 'Set-ItemProperty' in expression or statement.
    + CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
    + FullyQualifiedErrorId : MissingExpressionAfterOperator

PS C:\WINDOWS\system32> + ~~~~~~~~~~~~~~~~
```

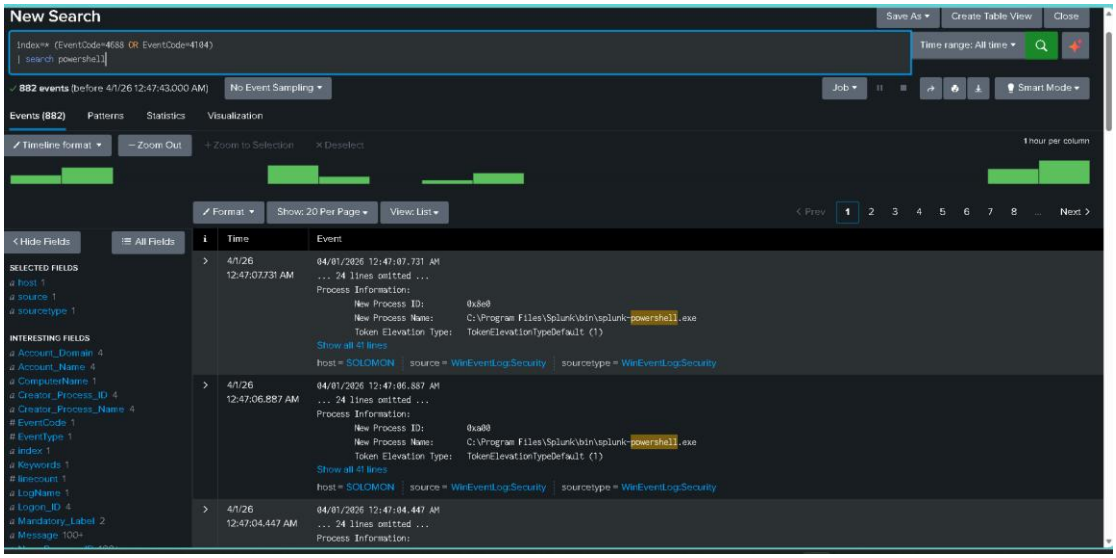


Figure 6: Detection of PowerShell Activity via Command Analysis

c. Suspicious Process Execution

Command-line tools such as cmd.exe and PowerShell were executed to mimic were executed to simulate attacked behavior.

Figure 7: Execution of whoami in cmd

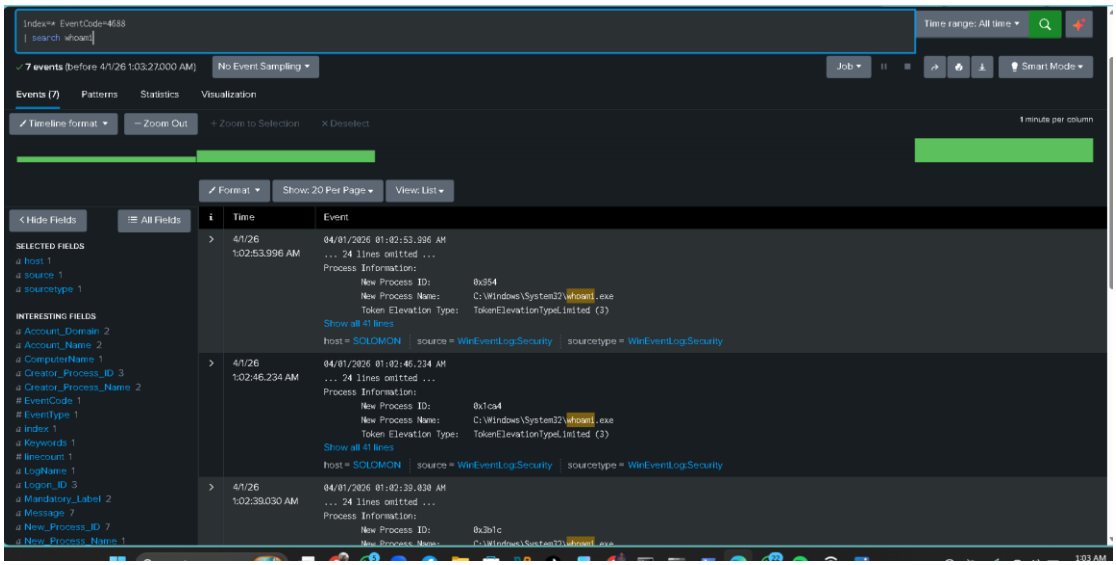


Figure 8: Detection of cmd activity via Command Analysis

RESULTS AND ANALYSIS:

The SIEM system successfully detected all simulated attacks.

- Brute force attacks were identified through repeated failed login attempts
- PowerShell misuse was detected via command-line monitoring
- Suspicious processes were tracked using Sysmon logs
- Network activity was visualized and analyzed

The dashboard provided real-time insights into system activities, improving threat visibility and response capability.

DASHBOARD IMPLEMENTATION:

A monitoring dashboard was created in Splunk to display:

- Failed login attempts over time
- PowerShell execution trends
- Top executed processes
- Network connections

This dashboard enabled centralized monitoring and quick identification of anomalies.

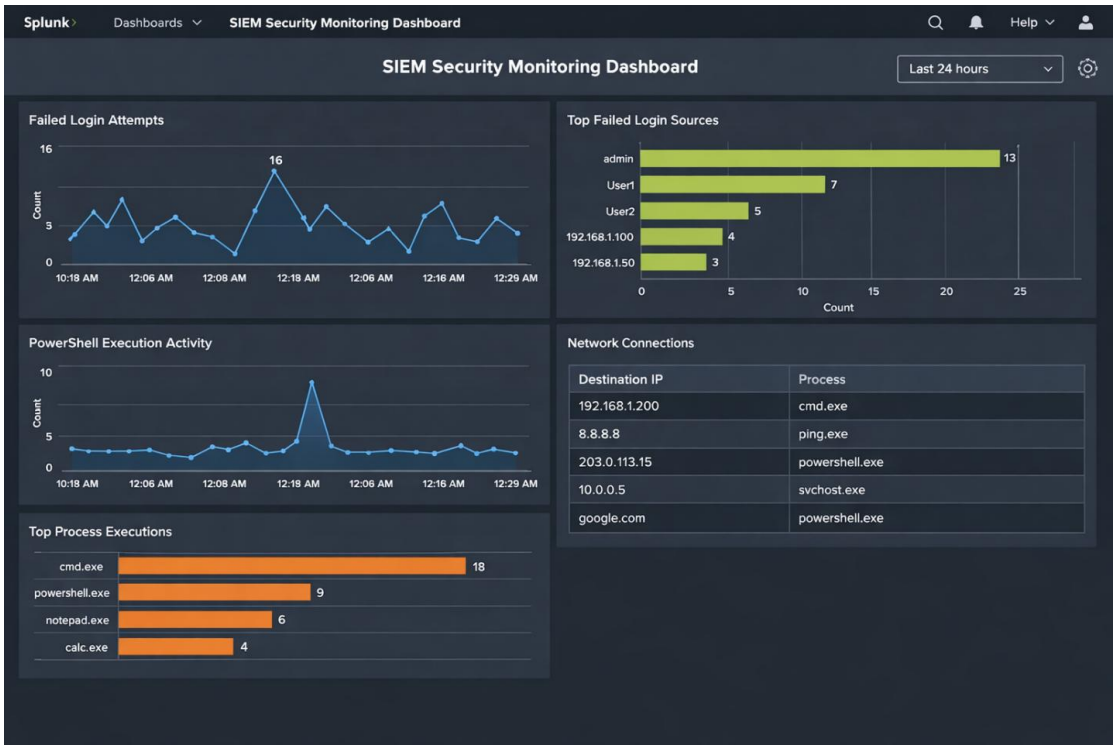


Figure 9: SIEM Dashboard for Real-Time Monitoring of Security Events

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CHALLENGES:

- Initial configuration of log forwarding
- Ensuring Sysmon logs were properly indexed
- Filtering relevant events from large datasets

CONCLUSION

This project demonstrates how SIEM systems can detect and analyze cyber threats using real-time log monitoring and correlation. By integrating Splunk with Sysmon, a robust monitoring environment was created that simulates real-world Security Operations Center (SOC) activities. The implementation highlights the importance of log analysis and detection engineering in cybersecurity operations.